

QCI Central Finite Curve

Production Implementation Plan — IT Evaluation Brief

Background & Purpose

Central Finite Curve is a RAG-powered document generation platform being built for QCI. It ingests institutional documents (Work Orders, Proposals, Reports) and enables staff to generate new drafts grounded in prior work. A working demo was validated on 23 April 2026 with 24 documents on Railway PaaS.

This brief is for IT team evaluation. Three infrastructure options are presented below. IT is requested to assess compliance feasibility, flag any blockers (especially on data residency and VPN), and provide inputs before senior leadership review.

Demo vs. Production Gap

The demo environment is not representative of production scale:

Component	Demo State	Production Requirement
Documents	24 PDFs (~50 MB)	~80 GB corpus (thousands of docs)
Concurrent Users	1–3	~500
Auth	Shared password + JWT	SSO / RBAC
OCR	RunPulse free tier	Enterprise / Docling hybrid
Database	Railway Postgres (basic)	Managed, HA, with backups
Storage	Railway volume mount	S3-compatible blob storage (India region)
LLM	Gemini Flash free tier	Paid tier or alternative

Key Technical Decision Areas — IT Input Required

The following areas require IT sign-off before implementation begins:

Area	Core Question	Direction
OCR	RunPulse enterprise vs open-source?	Docling (bulk) + RunPulse for new docs only
Hosting	Railway vs Azure/AWS?	Azure Central India (data residency)
Database	Railway Postgres vs managed?	Azure PostgreSQL Flexible Server + pgvector
Security / VPN	How to ingest QCI server docs safely?	Site-to-site VPN, read-only mount, audit logs
LLM	Paid vs free tier?	Gemini Flash paid tier (<₹5K/mo at scale)
Dev Tooling	AI coding access?	Cursor Ultra (\$200/mo) — all models, full agents

DATA RESIDENCY: QCI is a government body. Railway hosts data on US/EU servers only. If QCI IT security policy requires data to remain in India, Option A is disqualified and Option B (Azure Central India) or Option C (AWS Mumbai) is mandatory.

VPN (CRITICAL PATH): Ongoing document ingestion from QCI's local server requires a site-to-site VPN configured by QCI IT (read-only mount). This is on the critical path for all options. A manual upload interim is available while VPN is set up.

Options A, B, and C follow on the next three pages

Option A — Lean Production (Railway + Incremental Hardening)

"Ship fast, harden later"

Stay on Railway Enterprise, add proper auth and monitoring. Fastest and cheapest path. Data residency compliance must be confirmed by IT before selecting this option.

Architecture

Component	Detail
Hosting	Railway Pro/Enterprise — currently in use; upgrade to Enterprise plan
Database	Railway Postgres with backups enabled
OCR	Docling open-source (bulk ingest) + RunPulse (new documents only)
LLM	Gemini Flash (paid tier)
Auth	JWT with user accounts (no SSO in this phase)
VPN	WireGuard tunnel to QCI local server (QCI IT to configure)
Frontend	Vercel free tier or Railway static hosting
Dev Tooling	Cursor Ultra (\$200/mo) — all frontier models, full agent mode

Budget

Category	Best Case	Normal Case	Worst Case
Railway hosting	₹5K/mo	₹12K/mo	₹25K/mo
RunPulse OCR (ongoing)	₹15K/mo	₹40K/mo	₹80K/mo
Gemini API (LLM generation)	₹500/mo	₹2K/mo	₹5K/mo
Cursor Ultra (dev tooling)	₹17K/mo	₹17K/mo	₹17K/mo
Domain + SSL	₹500/mo	₹500/mo	₹500/mo
Monthly Total	₹38K	₹71.5K	₹1.27L
One-time setup	₹50K	₹1L	₹2L
Year 1 Total	₹5.1L	₹9.6L	₹17.2L

Trade-offs

Advantages

- Fastest path to production: 4–6 weeks
- Lowest one-time setup cost
- Buildable with Cursor Ultra; QCI IT involvement minimal (VPN + domain only)

Limitations

- Data hosted on US/EU servers — may fail QCI IT security or compliance audit
- No SSO or enterprise compliance features in initial phase
- Prabhav cannot run DARPG work in parallel (for the build phase)
- ⚠ IT team to verify Railway data residency compliance before selecting this option

Timeline	4–6 weeks (best) · 6–8 weeks (worst)
Staffing	Prabhav (primary developer) + QCI IT (VPN setup and domain pointing only)

Option B — Azure India Hybrid ★ Recommended

"Government-ready without over-engineering"

Move compute and database to Azure Central India for data residency. Use managed Azure services to minimise operational burden. Integrates with QCI's existing Microsoft stack.

Architecture

Component	Detail
Hosting	Azure App Service (B2 plan, 2 instances) — Central India region
Database	Azure Database for PostgreSQL Flexible Server + pgvector extension
OCR	Docling open-source (bulk ingest) + Azure Document Intelligence (new docs)
LLM	Gemini Flash via API (or Azure OpenAI for full data-in-Azure compliance)
Auth	Azure AD B2C / QCI's existing SSO infrastructure
VPN	Azure VPN Gateway → QCI local server (managed, enterprise-grade)
Storage	Azure Blob Storage (India region)
Dev Tooling	Cursor Ultra (\$200/mo) — all frontier models, full agent mode

Budget

Category	Best Case	Normal Case	Worst Case
Azure Compute (App Service)	₹8K/mo	₹18K/mo	₹35K/mo
Azure PostgreSQL Flexible Server	₹5K/mo	₹10K/mo	₹20K/mo
Azure Blob Storage	₹500/mo	₹1.5K/mo	₹3K/mo
Azure VPN Gateway	₹3K/mo	₹3K/mo	₹5K/mo
OCR (Azure Document Intelligence)	₹2K/mo	₹5K/mo	₹15K/mo
LLM API (Gemini / OpenAI)	₹500/mo	₹3K/mo	₹8K/mo
Cursor Ultra (dev tooling)	₹17K/mo	₹17K/mo	₹17K/mo
Monitoring + misc.	₹1K/mo	₹2K/mo	₹5K/mo
Monthly Total	₹37K	₹59.5K	₹1.08L
One-time setup	₹1.5L	₹3L	₹5L
Year 1 Total	₹5.9L	₹10.1L	₹18L

Trade-offs

Advantages

- Data residency in India — meets government compliance requirements
- Azure AD integration enables SSO with QCI's existing infrastructure
- Managed services minimise ongoing maintenance overhead
- Azure Hybrid Benefit (if QCI has M365 licensing) can reduce compute costs 30–40%

Limitations

- More setup complexity than Railway (8–12 weeks vs 4–6 weeks)
- Prabhav cannot run DARPG work in parallel during Azure buildout phase
- ⚠ Optional: Azure consultant for 2–3 weeks (₹50K–1L) would significantly accelerate initial setup

Timeline	8–10 weeks (best) · 12–14 weeks (worst)
Staffing	Prabhav (primary, ~80% of build) + QCI IT (Azure subscription, VPN, AD integration, security review) + optional Azure consultant (2–3 weeks part-time)

Option C — Full Enterprise (Dedicated Infrastructure + Ops Team)

"Maximum control, maximum cost"

Full cloud deployment on Kubernetes with a dedicated DevOps engineer, CI/CD pipeline, staging environment, self-hosted OCR, and comprehensive monitoring. Included for completeness — represents production-grade infrastructure for a large government IT deployment.

Architecture

Component	Detail
Hosting	Azure AKS (Kubernetes) or AWS ECS — full container orchestration
Database	Azure PostgreSQL HA (zone-redundant, automated failover)
OCR	Self-hosted Docling on dedicated GPU instances
LLM	Azure OpenAI (data stays entirely within Azure boundary)
Auth	Full SSO with RBAC, MFA, and conditional access policies
CI/CD	GitHub Actions → staging environment → production pipeline
Monitoring	Azure Monitor + Application Insights + alerting rules
Security	External security audit, penetration testing, WAF

Budget

Category	Best Case	Normal Case	Worst Case
Azure Infrastructure (AKS/ECS)	₹40K/mo	₹80K/mo	₹1.5L/mo
DevOps Engineer (contract hire)	₹80K/mo	₹1.2L/mo	₹1.5L/mo
GPU Compute (self-hosted OCR)	₹15K/mo	₹30K/mo	₹50K/mo
LLM API (Azure OpenAI)	₹3K/mo	₹8K/mo	₹20K/mo
Monitoring + security tools	₹5K/mo	₹10K/mo	₹20K/mo
Monthly Total	₹1.43L	₹2.48L	₹3.9L
One-time setup	₹5L	₹10L	₹15L
Year 1 Total	₹22L	₹40L	₹62L

Trade-offs

Advantages

- Maximum security, compliance, and operational control
- Full high-availability and disaster recovery
- Production-grade CI/CD with staging environment
- Comprehensive monitoring and alerting infrastructure

Limitations

- Highest cost — 4–6× Option B Year 1 budget
- Requires a dedicated DevOps hire (cannot be built solo)
- Over-engineered for current ~500 user base
- Longest timeline: 16–24 weeks minimum

Timeline	16 weeks (best) · 20–24 weeks (worst)
Staffing	Prabhav + Dedicated DevOps Engineer (full-time) + QCI IT (active) + External Security Auditor

★ **IT Evaluation Requested:** Data residency policy applicability · Azure subscription provisioning path · VPN feasibility with QCI network team · SSO / Azure AD integration scope · Security audit requirements. Please flag any blockers to Option B before leadership presentation.